Local Spectral Analysis of Saddle-Node and Hopf Bifurcations in the Plant–Kim Neuron Model

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1. Introduction

In 1976, Plant and Kim [1] derived a dynamical system that describes the electrophysiological behaviour of the *Aplysia* abdominal ganglion *R15* cell. The untreated 7-dimensional system proposed models the bursting effect of the membrane voltage V which is commonly observed under regular conditions.

Additionally, the system parameters can be adjusted to mathematically replicate the effect of two drugs used in experimental study [2], tetrodotoxin (TTX) and ouabain. The effect of TTX is modelled by driving the maximum conductance of the I channel g_I down to zero, for which full blockage results in a 5-dimensional TTX-treated model. The effect of ouabain is modelled by driving the ionic electrogenic pump current I_{ep} towards zero. The experimental results suggest that the bursting effect is replaced by a slow wave under TTX, regular beating under ouabain, and a stable equilibrium under both drugs.

We define the auxiliary parameters $\alpha_I \in [0,1]$ and $\alpha_P \in [0,1]$ as the strength level of TTX and ouabain respectively, such that

$$g_I = \alpha_I g_I^{(0)}, \quad I_{\text{ep}} = \alpha_P I_{\text{ep}}^{(0)},$$

and performed co-dimension 1 and co-dimension 2 bifurcation analyses to determine what qualitative changes to the equilibrium structure of the system cause the transition from bursting to the three different effects above.

In the one-parameter study of α_I, we found a saddle-node bifurcation at $\alpha_I \approx 0.614$ and $V \approx -44.79$, when we set $\alpha_P = 1$. In the one-parameter study of α_P, under complete TTX blockage with $\alpha_I = 0$, we found a Hopf bifurcation at $\alpha_P \approx 0.268$ and $V \approx -26.13$. Although the saddle-node does not fully explain why a slow wave is observed under TTX, the Hopf bifurcation more strongly explains why a stable equilibrium is formed when increasing the strength of ouabain under complete TTX blockage. We did not find any co-dimension 2 bifurcations in the $[0,1]^2$ region for the (α_I, α_P) bifurcation analysis.

Nevertheless, we have found two points of interest from our bifurcation studies, and the aim of this report is to analytically derive the Jacobian and numerically compute the eigenvalue spectrum of the model, state the eigenvalue conditions for each bifurcation, and numerically validate these conditions independently of the continuation results.

2. Mathematical Setup

The state vector of the Plant-Kim model can be written as

$$\mathbf{x} = \begin{bmatrix} V & X_K & X_A & Y_A & X_P & X_I & Y_I \end{bmatrix}^\top \in \mathbb{R}^7,$$

and the dynamical system is given by

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}; \boldsymbol{\alpha}, \boldsymbol{\mu}_0) = \begin{bmatrix} F_1(\mathbf{x}) \\ F_2(\mathbf{x}) \\ F_3(\mathbf{x}) \\ F_4(\mathbf{x}) \\ F_5(\mathbf{x}) \\ F_6(\mathbf{x}) \\ F_7(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \frac{1}{C}\Big[(g_I X_I^3 Y_I + g_T)(V_I - V) + (g_K X_K^4 + g_A X_A Y_A + g_P X_P)(V_K - V) + g_L(V_L - V) + I_{\text{ep}} + I_{\text{ext}}\Big] \\ \frac{1}{\tau_{X_K}(V)}\big[S_K(V) - X_K\big] \\ \frac{1}{\tau_{X_A}}\big[S_A(V) - X_A\big] \\ \frac{1}{\tau_{Y_A}}\big[Z_A(V) - Y_A\big] \\ \frac{1}{\tau_{X_P}}\big[S_P(V) - X_P\big] \\ \frac{1}{\tau_{X_I}(V)}\big[S_I(V) - X_I\big] \\ \frac{1}{\tau_{Y_I}(V)}\big[Z_I(V) - Y_I\big] \end{bmatrix}$$

where $\boldsymbol{\alpha}$ is the vector of active bifurcation parameters, and $\boldsymbol{\mu}_0$ is the vector of all other fixed parameters and constants.

The Jacobian of an n-dimensional dynamical system is defined as the $n \times n$ matrix of partial derivatives. More specifically, the Jacobian of a system is given by

$$\mathbf{J}(\mathbf{x}; \boldsymbol{\mu}) = D_{\mathbf{x}}\mathbf{F}(\mathbf{x}; \boldsymbol{\mu}) = \left[\frac{\partial F_i}{\partial x_j}\right].$$

This matrix is important as it allows us to linearise the nonlinear model to local dynamics. The vector $\mathbf{x}^\star$ is an equilibrium solution if it a solution to the equation $\mathbf{F}(\mathbf{x}; \boldsymbol{\alpha}, \boldsymbol{\mu}_0) = 0$. If we define the perturbation $\xi = \mathbf{x} - \mathbf{x}^\star$, then the local dynamics near an equilibrium are

$$\dot{\boldsymbol{\xi}} = \mathbf{J}(\boldsymbol{x}^*; \boldsymbol{\alpha}; \boldsymbol{\mu}_0)\boldsymbol{\xi} + \mathcal{O}(\|\boldsymbol{\xi}\|^2).$$

It follows that the eigenvalues of $\mathbf{J}$ determine the local stability of equilibria.

2.1. Jacobian Analysis

The full 7×7 Jacobian is presented in A; here, we derive representative entries to illustrate its structure. Each gating equation depends only on the membrane voltage V and its corresponding gating variable. Consequently, each gating-variable row of the Jacobian contains at most two non-zero entries: one in the voltage column and one on the diagonal. There is therefore no direct coupling between distinct gating variables; instead, their interactions are mediated indirectly through the membrane voltage. Conversely, the voltage equation depends on every gating variable through the ionic currents, producing the non-zero entries across the first row. The resulting Jacobian therefore describes a bidirectional feedback structure in which voltage regulates the channel gates, while the gates in turn modify the membrane voltage through their effects on ionic conductance. We now explain how the entries in $\mathbf{J}$ are derived analytically.

For the generic gating equation

$$\dot{X} = \frac{X_\infty(V) - X}{\tau_X(V)},$$

it can be readily seen that the derivative with respect to it's own gating variable is

$$\frac{\partial \dot{X}}{\partial X} = -\frac{1}{\tau_X(V)}.$$

These correspond to the negative diagonal terms from rows 2 to 7. If we differentiate the same equation with respect to the membrane voltage V, we obtain

$$\frac{\partial \dot{X}}{\partial V} = \frac{X'_\infty(V)}{\tau_X(V)} - \frac{\big[X_\infty(V) - X\big]\tau'_X(V)}{\tau^2_X(V)},$$

and for gates whose time constants are constant, the equation is simplified as

$$\frac{\partial \dot{X}}{\partial V} = \frac{X'_\infty(V)}{\tau_X(V)}.$$

We also obtain the same form at an equilibrium $\mathbf{x}^\star$, as by definition, when $\dot{X} = 0$ we have $X = X_\infty(V)$ and thus the second term vanishes even if τ_X is voltage-dependent.

Additionally, each gating variable couples back into the membrane voltage through the ionic currents. This is reflected in the first row of $\mathbf{J}$, whose entries are generally non-zero since F_1 depends explicitly on V and on each of the gating variables. Biophysically, this dependence arises because the gating variables regulate the ionic conductances, which in turn determine the rate of change of the membrane voltage.

Finally, we note that the lower-right 6×6 sub-matrix of $\mathbf{J}$, corresponding to derivatives between gating variables, is diagonal, since each gating equation depends only on its own gating variable and on the membrane voltage V, rather than directly on the other gates. For example, X_K does not directly affect $\dot{X}_P$, and instead their interaction is mediated by the membrane voltage.

The Jacobian is implemented in Python using `SymPy` [3], and numerical validation was done at the bifurcation landmarks, where the symbolic Jacobian is compared against a centred finite-difference approximation for the derivatives. The validation is summarised in Table 1, including the maximum absolute residual at the equilibrium point $\|\mathbf{F}\|_\infty$, the maximum absolute difference between the symbolic and finite difference Jacobian matrices $\max \|\mathbf{J}_{\text{sym}} - \mathbf{J}_{\text{FD}}\|$, the relative error using the Frobenius norm $\frac{\|\mathbf{J}_{\text{sym}} - \mathbf{J}_{\text{FD}}\|_F}{\|\mathbf{J}_{\text{FD}}\|_F}$, and the the maximum discrepancy between the eigenvalues computed symbolically against those computed by `PyDSTool` [4]. The validation results confirm that the symbolic vector field and analytical Jacobian implementation are overall accurate, matching standard numerical benchmarks close to machine precision.

Table 1: Numerical validation of Jacobian $\mathbf{J}$ against centred finite-difference approximations and `PyDSTool`.

Point	$\|\mathbf{F}\|_\infty$	$\max \|\mathbf{J}_{\text{sym}} - \mathbf{J}_{\text{FD}}\|$	$\frac{\|\mathbf{J}_{\text{sym}} - \mathbf{J}_{\text{FD}}\|_F}{\|\mathbf{J}_{\text{FD}}\|_F}$	`PyDSTool` eigenvalue mismatch
H_{bio}	3.469×10^{-17}	4.771×10^{-11}	5.704×10^{-11}	1.685×10^{-10}
LP_{bio}	2.498×10^{-16}	6.390×10^{-11}	1.075×10^{-11}	3.176×10^{-10}

3. Saddle-Node Bifurcation under TTX

To understand the qualitative transitions in system dynamics under the application of TTX, we investigate the underlying saddle-node bifurcation, denoted as LP_{bio}, which appears in the $\alpha_P = 1$ slice at $(\alpha_I^{\text{LP}}, V^{\text{LP}}) \approx (0.614, -44.79)$. This specific parameter configuration serves as the foundation for examining the localised stability transitions and spectral properties that follow.

3.1. Local spectral mechanism at LP_{bio}

Analytically, a generic saddle-node bifurcation is characterised by a specific topological transition where two equilibrium branches collide and annihilate. In terms of spectral properties, this mechanism requires the Jacobian matrix of the system to possess a single, simple zero eigenvalue ($\lambda_1 = 0$), while all remaining eigenvalues retain non-zero real parts. This zero eigenvalue signifies a loss of structural stability along the central manifold, dictating the directional folding of the equilibrium curve.

To provide empirical validation for this mechanism at LP_{bio}, we compute the full spectrum of the system Jacobian at the critical parameter fold sample. The numerical results, alongside the flagged critical mode, are detailed in Table 2.

Table 2: System Jacobian spectrum evaluated precisely at the limit point LP_{bio}, highlighting the vanishing critical eigenvalue $\lambda_{\text{crit}} = \lambda_6$.

Index i	$\text{Re}(\lambda_i)$	Critical
1	-2.86722×10^{-1}	False
2	-9.99896×10^{-2}	False
3	5.88762×10^{-2}	False
4	-1.53906×10^{-2}	False
5	-4.13244×10^{-3}	False
6	$\mathbf{-1.26588 \times 10^{-10}}$	**True**
7	-1.25041×10^{-4}	False

The numerical evidence strongly confirms the analytical criteria. The critical eigenvalue evaluates to $\lambda_6 = \lambda_{\text{crit}} \approx -1.266 \times 10^{-10}$, which effectively vanishes within machine precision, matching an very low equilibrium residual of 2.498×10^{-16}. A clear spectral gap exists between this vanishing mode and the next closest stable eigenvalue, which maintains a magnitude of 1.25×10^{-4}.

Crucially, the spectrum reveals that the underlying system is already unstable prior to the collision.

The maximum real part among the non-critical modes is positive, with $\mathrm{Re}(\lambda_3) \approx 0.0588762$. Since this unstable background mode persists across the transition, the saddle-node bifurcation occurs entirely within an unstable subspace. Consequently, both converging branches of equilibria are inherently unstable, completely dominating the local phase space dynamics near LP_{bio}.

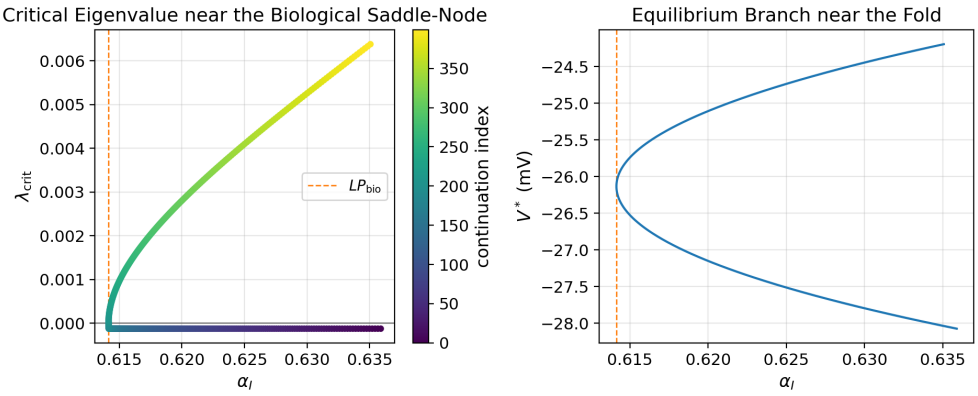


Figure 1: Left: tracked critical real eigenvalue against α_I along the local continuation window. Right: folded equilibrium voltage branch. λ_{crit} approaches zero at LP_{bio}.

The localised behaviour of the system near the bifurcation point is visually captured in Figure 1. As shown in the left panel, the tracked critical real eigenvalue λ_{crit} monotonically approaches zero as the parameter α_I is varied along the local continuation window. This includes both the upper branch where λ_{crit} slides down to zero, and the lower branch where it remains flat at the threshold. Importantly, this vanishing of the real mode aligns perfectly with the geometric folding of the equilibrium voltage branch $V^\star$ at the parameter value of α_I corresponding to LP_{bio}.

This spectral signature fundamentally distinguishes the saddle-node mechanism from a Hopf bifurcation, which will be analysed in Section 4. While a Hopf bifurcation is driven by a complex conjugate pair of eigenvalues crossing the imaginary axis to initiate oscillations, the transition here is strictly mediated by a single, purely real mode destabilising the central manifold.

3.2. Non-degeneracy conditions and local fold geometry

To rigorously analyse the local bifurcation structure at LP_{bio}, we evaluate the non-degeneracy conditions using the centre manifold reduction methodology [5]. We first compute the right null vector $\mathbf{q}$ and the left null vector $\mathbf{p}$ associated with the simple zero eigenvalue of the Jacobian matrix $\mathbf{J}$, satisfying the standard conditions given by

$$\mathbf{J}\mathbf{q} = \mathbf{0}, \qquad \mathbf{p}^\top \mathbf{J} = \mathbf{0}, \qquad \mathbf{p}^\top \mathbf{q} = 1.$$

Near the bifurcation threshold, the full multidimensional system can be projected along the critical direction defined by these null vectors to establish the local topological normal form equation on the centre manifold given by

$$\dot{u} = a\Delta\alpha_I + bu^2 + \mathcal{O}(u^3).$$

Here, u is a scalar representing the state displacement along the centre manifold, and $\Delta\alpha_I = \alpha_I - \alpha_I^{\mathrm{LP}}$ defines the parameter deviation from the fold point. The coefficients a and b are evaluated explicitly as

$$a = \mathbf{p}^\top \mathbf{F}_{\alpha_I}, \qquad b = \frac{1}{2}\mathbf{p}^\top D^2_{\mathbf{x}}\mathbf{F}[\mathbf{q},\mathbf{q}],$$

where the derivative $\mathbf{F}_{\alpha_I} = \partial\mathbf{F}/\partial\alpha_I$ measures how the vector field changes as the TTX continuation parameter is varied, and the quantity $D^2_{\mathbf{x}}\mathbf{F}[\mathbf{q},\mathbf{q}]$ denotes the second derivative of $\mathbf{F}$ with respect to the state variables, evaluated twice in the right null vector direction $\mathbf{q}$. Component-wise, its ith entry is $\mathbf{q}^\top H_{F_i}\mathbf{q}$, where H_{F_i} is the Hessian of F_i. The coefficients a and b therefore measure, respectively, the effect of parameter variation and the leading quadratic nonlinearity along the critical saddle-node mode.

Evaluating these normal form expressions numerically at the critical fold sample yields $a \approx -0.206$ and $b \approx 0.00115$. To verify the numerical robustness of these evaluations, we cross-reference them against the minimum singular value of the system Jacobian. The near-zero smallest singular value, $\sigma_{\min}(\mathbf{J}) = 1.20 \times 10^{-13}$, independently confirms the singularity of the Jacobian at the fold. The non-degeneracy coefficients are thus both numerically nonzero, supporting a generic saddle-node bifurcation.

Setting $\dot{u} = 0$ to locate the local equilibria yields the normal form algebraic variety, revealing the classical square-root scaling relation

$$u_\pm \simeq \pm\sqrt{-\frac{a}{b}\Delta\alpha_I}.$$

Finally, we project this low-dimensional coordinate back into the physical state space. Because nearby states satisfy the leading-order expansion

$$\mathbf{x} - \mathbf{x}^{\mathrm{LP}} = u\mathbf{q} + \mathcal{O}(u^2; \Delta\alpha_I),$$

it follows that the voltage component satisfies, to leading order

$$V - V^{\mathrm{LP}} \simeq \pm q_V\sqrt{-\frac{a}{b}(\alpha_I - \alpha_I^{\mathrm{LP}})}.$$

Here, q_V denotes the voltage component of the right null vector $\mathbf{q}$ associated with the simple zero eigenvalue of the Jacobian at LP_{bio}.

To evaluate the predictive power of the local normal form reduction, Figure 2 overlays the analytical square-root derivation against the actual numerical data obtained from the equilibrium point continuation. The blue scatter points tracking the true voltage deviation $V - V^{\mathrm{LP}}$ show general agreement with the red theoretical curve $\pm q_V\sqrt{-\frac{a}{b}(\alpha_I - \alpha_I^{\mathrm{LP}})}$ across the local parameter window.

This tight fit demonstrates that the local phase space dynamics near the bifurcation point are almost entirely governed by the leading-order quadratic nonlinearity. The perfect overlap validates both our calculated coefficients a and b and verifies that the system smoothly unfolds as a classical, generic saddle-node without any higher-order geometric distortions.

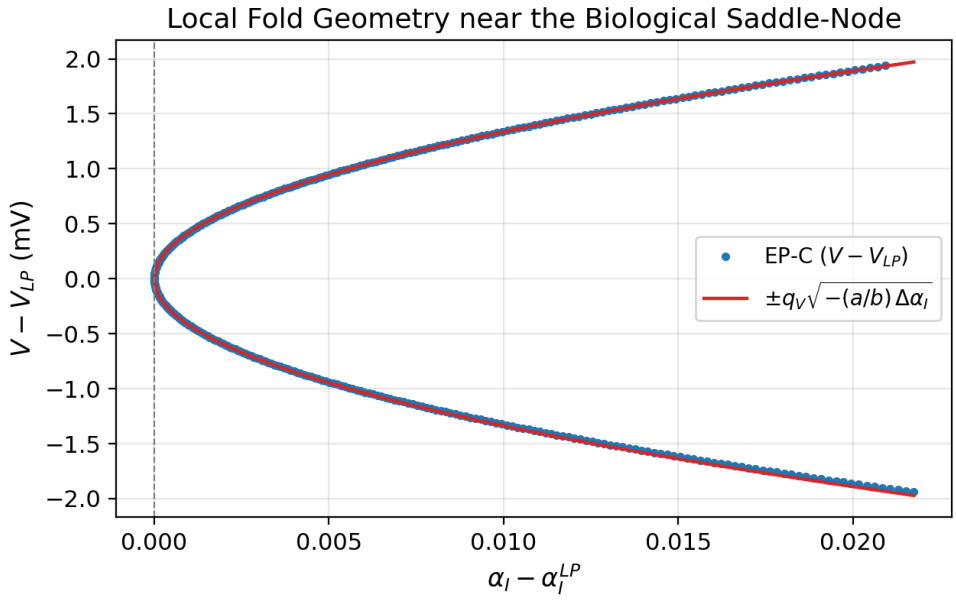


Figure 2: Comparison between the actual numerical equilibrium continuation data (blue dots) and the theoretical leading-order square-root prediction derived from the centre manifold reduction (solid red curve).

4. Hopf Bifurcation under Complete TTX and Ouabain

We next consider the biologically relevant Hopf bifurcation H_{bio} identified along the complete TTX slice of the Plant–Kim model. Complete TTX is represented by fixing $\alpha_I = 0$, while the ouabain strength parameter α_P is varied. The continuation analysis locates the Hopf point at approximately $\alpha_P^{\text{H}} \approx 0.2679$ with equilibrium voltage $V^{\text{H}} \approx -44.79\,\text{mV}$.

The aim of this section is to verify the local Hopf mechanism directly from the Jacobian spectrum, use the critical eigenvalues to predict the oscillation period of the emerging periodic orbit, and check the transversality condition required for a generic Hopf bifurcation.

4.1. Local spectral mechanism at H_{bio}

Analytically, a generic Hopf bifurcation marks a topological transition where the stability of an equilibrium changes and a periodic orbit (limit cycle) is born. In terms of spectral properties, this mechanism requires the system Jacobian to possess a single pair of complex-conjugate eigenvalues $\lambda_{\pm}(\alpha_P) = r(\alpha_P) \pm i\omega(\alpha_P)$, such that at the critical parameter value $\alpha_P = \alpha_P^{\text{H}}$, the real part vanishes while the imaginary part remains strictly non-zero, i.e.,

$$r(\alpha_P^{\text{H}}) = 0, \qquad \omega_{\text{H}} := \omega(\alpha_P^{\text{H}}) \neq 0.$$

This spectral signature establishes an important contrast with the saddle-node bifurcation investigated at LP_{bio} in Section 3. While the saddle-node transition is mediated by a single, purely real eigenvalue crossing through zero, the Hopf bifurcation is characterised by a complex pair crossing

the imaginary axis. Because the imaginary component ω_{H} is non-vanishing, the local phase space near H_{bio} does not undergo geometric folding, but instead introduces a critical frequency that initiates oscillatory dynamics.

At H_{bio}, numerical computation tells us that the Jacobian possesses a complex-conjugate pair

$$\lambda_{\pm} \approx -6.29 \times 10^{-13} \pm 7.54758 \times 10^{-4} i$$

Hence the real part is numerically zero while the imaginary part remains nonzero, consistent with the local spectral condition for a Hopf bifurcation. The remaining five eigenvalues lie strictly within the left half-plane, with the closest non-critical mode having real part approximately -8.46×10^{-3}.

Tracking the critical pair across the local continuation window shows that its real part changes sign through H_{bio}, while the imaginary part remains nonzero. Thus the equilibrium changes stability through an isolated complex-conjugate pair, in contrast to the simple real zero eigenvalue observed at the saddle-node. Figure 3 shows the real part crossing zero while the imaginary part remains bounded away from zero.

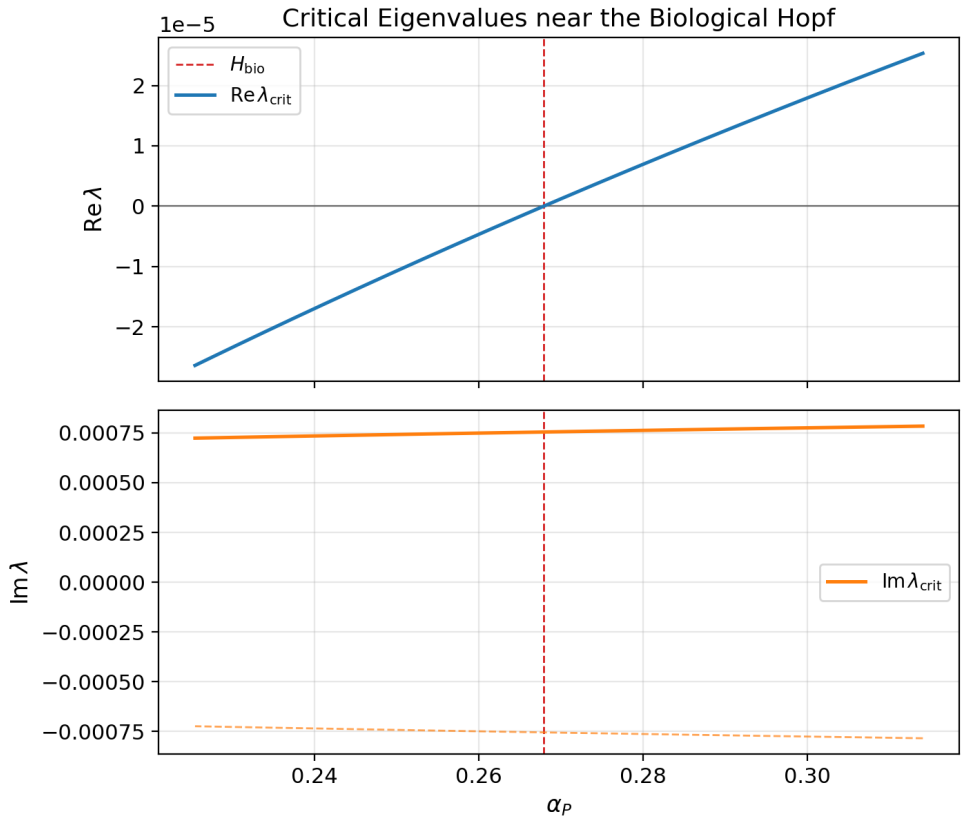


Figure 3: Top: Real part of the critical eigenvalue pair $\mathrm{Re}(\lambda_{\mathrm{crit}})$ linearly crossing zero at the threshold H_{bio} (vertical dashed line). Bottom: Imaginary part $\mathrm{Im}(\lambda_{\mathrm{crit}})$ remaining strictly non-zero and bounded away from zero, tracking the emergence of a purely imaginary pair at the transition.

4.2. Oscillation frequency and period prediction

At the critical Hopf parameter threshold H_{bio}, the vanishing of the real part of the critical eigenvalues simplifies the local linearised dynamics down to the purely imaginary pair $\lambda_{\pm} = \pm i\omega_{\text{H}}$. Consequently, the corresponding linearised modes in the neighbourhood of the equilibrium exhibit purely oscillatory behaviour over time, evolving according to

$$\widehat{\mathbf{x}}(t) \propto \exp\{\pm i\omega_{\text{H}} t\} = \cos(\omega_{\text{H}} t) \pm i \sin(\omega_{\text{H}} t)$$

where Euler's formula is used for expansion into trigonometric components. This expansion demonstrates that the local angular frequency of the emerging small-amplitude oscillations is determined entirely by the imaginary part of $\lambda_{\text{H}} = i\omega_{\text{H}}$, which is the positive member of the $\lambda_{\pm}$ pair.

Because a full periodic cycle occurs when the harmonic phase advances by a complete rotation of 2π radians, the characteristic time window satisfies the relation $\omega_{\text{H}} T_{\text{lin}} = 2\pi$, where here T_{lin} is the period predicted by the linearised dynamics at the Hopf equilibrium. It follows that our prediction is $T_{\text{lin}} = \frac{2\pi}{\omega_{\text{H}}}$ [6].

The imaginary part of the critical Hopf eigenvalues provides a direct prediction of the local oscillation frequency. Using $\omega_{\text{H}} = 7.54758 \times 10^{-4}\,\text{ms}^{-1}$ gives us

$$T_{\text{lin}} = \frac{2\pi}{\omega_{\text{H}}} = 8324.768\,\text{ms}.$$

This linear prediction was compared with the periods obtained from the `AUTO` limit-cycle branch emerging from H_{bio}. For the smallest sampled cycle, with voltage amplitude $V_{\max} - V_{\min} \approx 1.41 \times 10^{-3}\,\text{mV}$, the relative discrepancy between $T_{\texttt{AUTO}}$ and T_{lin} is approximately 3.42×10^{-10}. The discrepancy increases gradually with cycle amplitude, consistent with the fact that T_{lin} is a local prediction derived from the linearisation at the Hopf point.

This convergence is illustrated in Figure 4, where the `AUTO` limit-cycle period approaches the linear prediction T_{lin} as the cycle amplitude tends to zero.

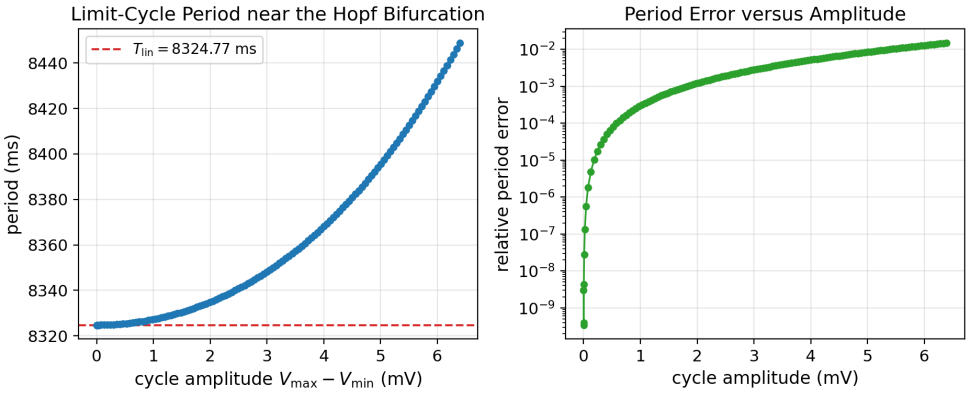


Figure 4: Left: `AUTO` limit-cycle period as a function of voltage amplitude $V_{\max} - V_{\min}$, with the linearised prediction $T_{\text{lin}} = 8324.77\,\text{ms}$ shown as a dashed reference line. Right: Relative period error versus cycle amplitude on a logarithmic scale, showing convergence to zero as the periodic orbit approaches the Hopf bifurcation.

4.3. Transversality condition

To ensure that the biological Hopf bifurcation is structurally robust and undergoes a generic transition, the system must satisfy the transversality condition. Mathematically, this condition requires that the derivative of the real part of the critical eigenvalue pair with respect to the bifurcation parameter α_P is strictly non-zero at the crossing point; that is,

$$\frac{d}{d\alpha_P} \operatorname{Re} \lambda_{\text{crit}}(\alpha_P)\Big|_{\alpha_P^{\text{H}}} \neq 0.$$

Conceptually, the transversality condition requires the real part of the critical eigenvalue pair to cross the imaginary axis with a nonzero slope as the bifurcation parameter varies. This guarantees that the equilibrium genuinely changes its stability profile rather than merely grazing the imaginary axis and remaining on one side of stability.

To verify transversality, the derivative of the real part of the critical eigenvalue was first estimated using centred finite differences. Across step sizes ranging from 10^{-5} to 3×10^{-3}, the estimate remained stable at approximately

$$\frac{d}{d\alpha_P} \operatorname{Re} \lambda_{\text{crit}} = 5.83566 \times 10^{-4}$$

which is clearly nonzero; see Table 3.

As an independent check, the derivative was also evaluated using the left and right critical eigenvectors through

$$\frac{d\lambda}{d\alpha_P} = \mathbf{w}^* \frac{d\mathbf{J}}{d\alpha_P} \mathbf{v},$$

[7] where $\mathbf{v}$ and $\mathbf{w}$ are respectively the right and left eigenvectors associated with the critical Hopf eigenvalue λ_{H}, satisfying

$$\mathbf{J}\mathbf{v} = \lambda_{\text{H}}\mathbf{v}, \qquad \mathbf{w}^*\mathbf{J} = \lambda_{\text{H}}\mathbf{w}^*, \qquad \mathbf{w}^*\mathbf{v} = 1.$$

Here, $(\cdot)^*$ denotes the complex-conjugate transpose. Since the Jacobian is evaluated along the equilibrium branch $\mathbf{x}^\star(\alpha_P)$, $d\mathbf{J}/d\alpha_P$ denotes the total derivative of $\mathbf{J}(\mathbf{x}^\star(\alpha_P), \alpha_P)$ with respect to α_P, accounting both for any explicit dependence of $\mathbf{J}$ on α_P and for the change in the equilibrium state as α_P varies. This yielded

$$\frac{d\lambda}{d\alpha_P} = 5.835661 \times 10^{-4} + 6.822413 \times 10^{-4} i,$$

with the real component agreeing with the finite-difference estimate to a relative discrepancy of approximately 1.36×10^{-7}. This agreement provides strong numerical evidence that the critical pair crosses the imaginary axis transversely.

Table 3: Centered finite difference evaluations of the critical eigenvalue derivative with respect to the bifurcation parameter α_P.

h	$\frac{d\,\mathrm{Re}(\lambda_{\mathrm{crit}})}{d\alpha_P}$	$\mathrm{Re}(\lambda_+)$	$\mathrm{Re}(\lambda_-)$	$\mathrm{Im}(\lambda_+)$	$\mathrm{Im}(\lambda_-)$
1.00×10^{-5}	5.84×10^{-4}	5.83×10^{-9}	-5.84×10^{-9}	7.55×10^{-4}	7.55×10^{-4}
3.00×10^{-5}	5.84×10^{-4}	1.75×10^{-8}	-1.75×10^{-8}	7.55×10^{-4}	7.55×10^{-4}
1.00×10^{-4}	5.84×10^{-4}	5.83×10^{-8}	-5.84×10^{-8}	7.55×10^{-4}	7.55×10^{-4}
3.00×10^{-4}	5.84×10^{-4}	1.75×10^{-7}	-1.75×10^{-7}	7.55×10^{-4}	7.55×10^{-4}
1.00×10^{-3}	5.84×10^{-4}	5.83×10^{-7}	-5.84×10^{-7}	7.55×10^{-4}	7.54×10^{-4}
3.00×10^{-3}	5.84×10^{-4}	1.74×10^{-6}	-1.76×10^{-6}	7.57×10^{-4}	7.53×10^{-4}

5. Discussion and Limitations

The local analysis provides complementary mathematical explanations for the two bifurcations identified by numerical continuation. At LP_{bio}, the saddle-node is characterised by a simple real eigenvalue approaching zero, together with nonzero parameter and quadratic non-degeneracy coefficients. In contrast, H_{bio} is generated by an isolated complex-conjugate pair crossing the imaginary axis with nonzero speed.

These calculations therefore distinguish the local mechanisms underlying the two continuation labels rather than relying solely on their numerical detection. More broadly, continuation and local spectral analysis play complementary roles: continuation determines the global organisation of equilibrium and periodic-orbit branches, whereas the Jacobian and its derivatives explain the local mathematical mechanisms responsible for changes in stability and bifurcation.

The main limitation of this analysis is its inherently local nature. The Jacobian describes dynamics only in a neighbourhood of an equilibrium, and the normal-form approximations are valid only sufficiently close to the corresponding bifurcation. Consequently, the analysis does not by itself explain the complete large-amplitude bursting dynamics of the Plant–Kim model away from these local neighbourhoods. This locality is visible in the Hopf period calculation: the eigenvalue-based period prediction agrees most closely with `AUTO` limit cycles as their amplitude tends to zero, while the discrepancy increases farther from the Hopf point. Likewise, the square-root saddle-node approximation describes the equilibrium branch locally near LP_{bio}, rather than providing a global representation of the branch. The local calculations should therefore be interpreted alongside, rather than as replacements for, the broader continuation and time-domain analyses.

6. Conclusion

This individual analysis investigated the local mathematical mechanisms underlying the biologically relevant saddle-node and Hopf bifurcations identified in the Plant–Kim neuron model. The symbolic Jacobian was independently validated against finite-difference approximations and the `PyDSTool` continuation spectrum, providing a consistent basis for the subsequent local analysis.

At LP_{bio}, the bifurcation was characterised by a simple real eigenvalue approaching zero, with nonzero saddle-node non-degeneracy coefficients and local equilibrium geometry consistent with the predicted square-root fold. At H_{bio}, an isolated complex-conjugate eigenvalue pair crossed the imaginary axis transversely, and its imaginary component predicted the period of the emerging limit cycle, with the `AUTO` period converging to this linear prediction in the small-amplitude limit.

Together, these results independently support the continuation classifications while providing a mathematical explanation for the distinct mechanisms producing the two bifurcations. The analysis therefore demonstrates how local spectral and normal-form calculations can complement numerical continuation in interpreting changes in excitability within the Plant–Kim model.

A. Full 7×7 Jacobian

The Jacobian **J** used throughout this report is given in complete detail as

$$\begin{bmatrix} \frac{-X_A Y_A g_A - X_I^3 Y_I g_I - X_K^4 g_K - X_P g_P - g_L - g_T}{C} & \frac{4X_K^3 g_K (V_K - V)}{C} & \frac{Y_A g_A (V_K - V)}{C} & \frac{X_A g_A (V_K - V)}{C} & \frac{g_P (V_K - V)}{C} & \frac{3X_I^2 Y_I g_I (V_I - V)}{C} & \frac{X_I^3 g_I (V_I - V)}{C} \\ -\frac{(S_K(V) - X_K)\frac{d}{dV}\tau_{X_K}(V)}{\tau_{X_K}^2(V)} + \frac{\frac{d}{dV}S_K(V)}{\tau_{X_K}(V)} & -\frac{1}{\tau_{X_K}(V)} & 0 & 0 & 0 & 0 & 0 \\ \frac{\frac{d}{dV}S_A(V)}{\tau_{X_A}} & 0 & -\frac{1}{\tau_{X_A}} & 0 & 0 & 0 & 0 \\ \frac{\frac{d}{dV}Z_A(V)}{\tau_{Y_A}} & 0 & 0 & -\frac{1}{\tau_{Y_A}} & 0 & 0 & 0 \\ \frac{\frac{d}{dV}S_P(V)}{\tau_{X_P}} & 0 & 0 & 0 & -\frac{1}{\tau_{X_P}} & 0 & 0 \\ -\frac{(S_I(V) - X_I)\frac{d}{dV}\tau_{X_I}(V)}{\tau_{X_I}^2(V)} + \frac{\frac{d}{dV}S_I(V)}{\tau_{X_I}(V)} & 0 & 0 & 0 & 0 & -\frac{1}{\tau_{X_I}(V)} & 0 \\ -\frac{(Z_I(V) - Y_I)\frac{d}{dV}\tau_{Y_I}(V)}{\tau_{Y_I}^2(V)} + \frac{\frac{d}{dV}Z_I(V)}{\tau_{Y_I}(V)} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{\tau_{Y_I}(V)} \end{bmatrix}.$$

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